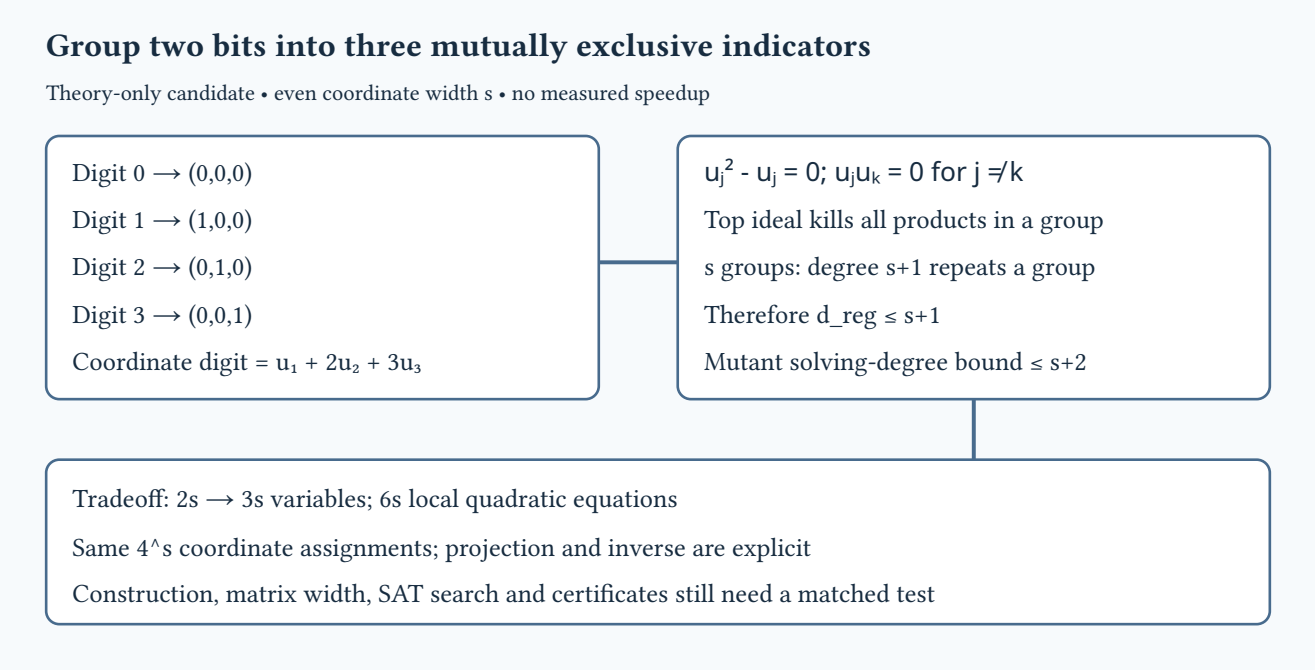
Grouped indicators: a structural regularity lead

Derived; independent review pending; no scientific trials

Represent every pair of coordinate bits by three mutually exclusive Boolean indicators. The all-zero indicator vector represents digit zero. Each original assignment has exactly one representation, and the coordinate remains affine in the new variables.



For two coordinates of even width s, the original S3 bit system has leading-form regularity s plus the ceiling of (s+2)/2 under the assumptions in the separate two-summand proof. The grouped system has 3s variables and 6s local quadratics, plus the substituted quartic. Its regularity is at most s+1: a monomial of degree s+1 repeats one of the s groups, so its within-group quadratic factor lies in the top ideal.

The inverse maps are explicit. For bits b0,b1, take u1=b0(1-b1), u2=(1-b0)b1, and u3=b0 b1. Conversely b0=u1+u3 and b1=u2+u3. Thus the result concerns an equivalent projected problem, with a changed variable count and degree filtration.

Salizzoni, Proposition 3.10, bounds the mutant-style solving degree by the maximum of regularity plus one and the maximum input degree. Here that gives s+2. This is an upper bound for the specified multiplication-closed-space convention. It does not prove a strict reduction in the actual solving degree of the original problem, or predict a particular software implementation's behavior.

The local top quotient has Hilbert series (1+3T) to the power s and total dimension 4 to the power s. The encoding retains all coordinate assignments. Lower degree therefore comes with wider graded components; runtime, memory and certificate costs remain unresolved. Fixed two-bit groups avoid the exponential membership-table construction of the other lead.

The full derivation, general group-width formula, proposed SAT interface, and separate unreviewed three-summand extension are in `next-leads.md`. Indicator encodings are established CSP tools; novelty has not been established. This page is outside the frozen two-summand membership proof and inherits no verdict from its review.